# CONSERVATION

**The negative impact that human beings** have on wildlife grows day by day, but so, too, does the impact of conservation. Across the world, organizations big and small are engaged in a concerted effort to protect nature in its original state, or to ensure that we use it in a sustainable way. It is a huge task, and one that raises some difficult practical and philosophical questions. Which is the best way of safeguarding species? How do you go about saving an animal that is on the verge of extinction? And, if resources are limited, are some animals more "important" than others? Experts do not always agree on the answers, but there is no doubt that conservation is an urgent priority if today's threatened species are to survive.

## Habitat protection

By far the most effective way of safeguarding animals is to protect their natural habitats. An animal's habitat provides everything necessary for its survival, and in its natural state it can continue to do this indefinitely as food and energy is passed from one species to another.

This is the thinking behind national parks and wildlife reserves. Even small parks can be effective—particularly when they protect breeding grounds—but, in general, the larger the area that is protected, the more species benefit and the greater are the chances that the habitat is truly self-sustaining. For example, Manú National Park—one of the largest in Peru—includes an extraordinary range of habitats from high-altitude Andean grassland to lowland Amazonian rainforest. It is home to more than 200 species of mammals, over 1,000 species of birds, and even more species of butterflies, making it one of the richest tropical reserves in the world. Its success is partly due to its remote location, which has restricted human settlement, unlike many parts of the Amazon farther east.

**WATERSIDE VANTAGE POINT**
*Specially constructed blinds allow visitors to watch birds in a wetland reserve. After centuries of drainage for agriculture, reserves like these are vitally important to many wetland species.*

In other parts of the world, national parks and reserves can suffer from their own popularity, and also from the pressure for resources. In the Galapagos Islands, for example, conservationists are engaged in an often difficult struggle to balance the needs of wildlife against the needs of an expanding human population, and increasing numbers of visitors.

## Techniques and technology

When a species is in immediate danger of extinction, captive breeding can be a highly effective way of bringing it back from the brink. In 1982, this was the situation with the California condor, when only about 24 birds were left in the wild. During the 1980s, a breeding program was initiated, and all the remaining birds were caught—a drastic measure that caused considerable controversy at the time. Three decades later, the intervention has been vindicated: the total population has reached over 430, with more than half this number flying free.

In recent years, new technology has played an increasing part in this kind of conservation work. Satellite tracking of released animals helps show where they feed and where they breed.

**READY FOR RELEASE**
*Raised in captivity, this California condor may one day help swell the population in the wild. However, in comparison with life in captivity, life in the wild can be difficult and even hazardous.*

Using DNA technology, there is even a possibility that recently extinct species could be "brought back to life". However, most conservationists believe that these techniques—on their own—are not long-term routes to survival. This is partly because they require a large commitment of time, money, and space. But a more significant problem lies in their outcome: if a species' natural habitat is disappearing, captive animals will have no home to go to if they are released.

## Controlling incomers

In isolated parts of the world, introduced, or "alien," species make life extremely difficult for native animals. Cats, foxes, and rats head the list of these problematic incomers, although plant-eating

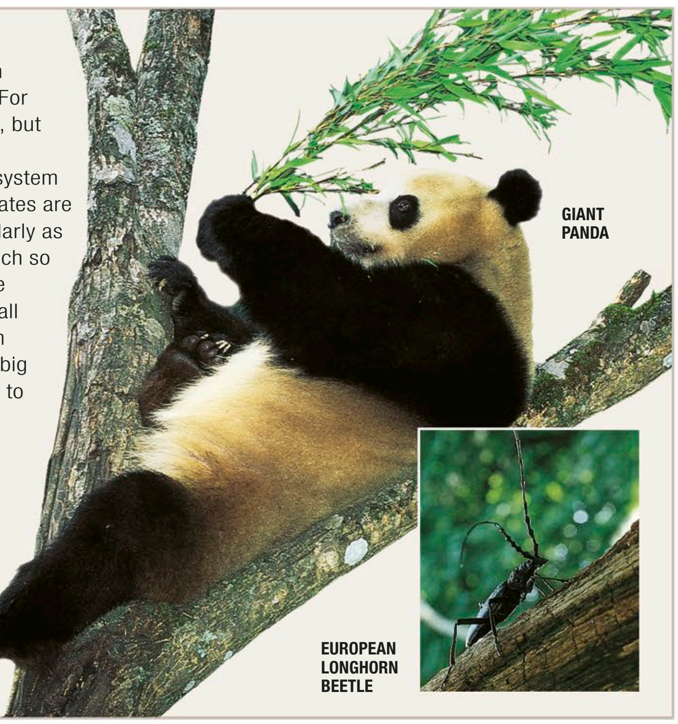
**EXCLUDING INTRUDERS**
*In Western Australia, this electric fence protects the Peron Peninsula from introduced mammals, such as cats. The entire peninsula—covering 390 square miles (1,000 square km)—is to become an "alien-free" haven for endangered marsupials.*

mammals can also cause immense damage. In some of the worst-affected regions, such as Australia and New Zealand, conservation programs are now under way to reduce this threat.

In an island as vast as Australia, eradicating feral cats or foxes is not a feasible goal. But in some parts of the country, large areas have been fenced off to protect bandicoots, bilbies, and other vulnerable marsupials. In these giant enclosures, alien species are either trapped or controlled by poison bait. The poisons are substances

### ANIMAL APPEAL

One problem with animal conservation is that our reactions to species differ. For example, everyone loves giant pandas, but far fewer like "creepy crawlies"—the invertebrates that underpin every ecosystem on land, and also in the sea. Invertebrates are essential for making life work, particularly as many of them recycle nutrients on which so many other living things depend. To be effective, conservation has to protect all animals in a habitat. These range from so-called "charismatic megafauna"—big animals with star-appeal—right down to animals that few people see, and that even fewer can name.




**POLES APART**
*The giant panda and the European longhorn beetle are at opposite ends of the spectrum of public interest and concern. While the panda attracts funds and media attention, the beetle and its like rarely arouse comment.*